

Forecasting Monthly Temperature Change in the ASEAN Region: A Data-Driven Framework for Climate Risk Preparedness and Regional Adaptation

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Abstract—ASEAN adaptation planning lacks comparable country-specific monthly temperature-change forecasts for near-term climate-risk preparedness. This study used a quantitative, data-driven, design-science research framework to develop and evaluate a tuned monthly temperature-forecasting pipeline for all 10 ASEAN member states. Using historical FAOSTAT Temperature Change on Land monthly series from 1961 to 2023, comprising 756 observations per country, missing historical values in Singapore were resolved using a leak-free regional Ridge Regression imputer. Four candidate models, Holt-Winters, SARIMA, Prophet, and XGBoost, were trained independently for each country and optimized through hyperparameter grid search using internal rolling-window cross-validation. Selected configurations were refitted on the 1961 to 2013 training history covering 636 months and evaluated on the 2014 to 2023 holdout test period covering 120 months using MAE, RMSE, epsilon-adjusted MAPE, sMAPE, and MASE. Prophet achieved the strongest overall regional performance by recording the lowest aggregate Mean Absolute Error of 0.50 degrees Celsius, Root Mean Squared Error of 0.62 degrees Celsius, sMAPE of 47.5 percent, and Mean Absolute Scaled Error of 0.98, though optimal model selection varied across national topographies. Statistical models achieved the highest accuracy in maritime equatorial countries, with forecasting errors below 0.30 °C. In contrast, Prophet and XGBoost performed better in mainland and monsoon-influenced countries by capturing broader temperature trends. These results suggest that selecting forecasting models based on country-specific climate characteristics can improve climate risk planning across ASEAN.

Keywords—ASEAN, Climate change adaptation, Grid search, Temperature anomalies, Time-series forecasting

I. INTRODUCTION

Southeast Asia faces increasing climate risk as rising temperatures intensify regional warming and physical climate hazards [1]. These changes affect human systems, adaptation needs, and climate-sensitive sectors such as health, agriculture, water resources, and disaster preparedness [2]. Observed records also show temperature-related extremes across Southeast Asia and nearby regions [3]. The risks are not uniform across ASEAN because regional climate exposure and hotspot patterns differ by location [4]. Island and coastal countries are affected by maritime and equatorial conditions, while mainland countries

are influenced by monsoon variability and larger continental temperature swings. These differences make monthly country-level temperature-change forecasts necessary for practical adaptation planning.

Adaptation offices need near-term forecasts that convert observed monthly temperature anomalies into planning information for individual countries [2]. A regional average alone cannot guide national decisions because it hides country-level differences in seasonality, trend, and forecast reliability.

Broad climate assessments establish regional warming, human-system impacts, observed Southeast Asian extremes, and hotspot differences [1], [2], [3], [4]. Forecasting studies show that statistical and machine-learning models vary across data structures, seasonality, and validation designs [5], [6], [7]. This study contributes to these domains by presenting a reproducible, standardized time-series benchmarking workflow applied systematically across all ten ASEAN member states, evaluating multiple model families under a single chronological framework, and assigning country-specific optimal models.

While previous climate studies have examined localized predictions or regional aggregates, there is a need for a unified comparative benchmark that evaluates statistical and machine-learning models under a single standardized chronological split, rolling-validation design, and unified accuracy protocol [5], [6], [7]. This study addresses this gap by presenting a comparable, country-specific benchmarking framework for all ten ASEAN countries, thereby enhancing the practical utility of historical anomaly records for near-term regional climate-risk preparedness.

This study addressed that gap by developing a tuned country-specific forecasting workflow. Specifically, it aimed to:

1. Characterize long-term monthly temperature-change patterns;
2. Evaluate Holt-Winters, SARIMA, Prophet, and XGBoost for each country;
3. Identify the best forecasting model per country; and

4. Interpret the results for adaptation-oriented planning.

II. METHODOLOGY

A. Research Design

The study used a quantitative, data-driven, design-science research design [8], [9]. The design produced a reproducible forecasting artifact for country-specific monthly temperature-change analysis, model tuning, final testing, and model assignment.

The workflow was organized into six connected stages: dataset filtering, preprocessing, missing-value treatment, model training, hyperparameter tuning, and final evaluation. The regional records were first separated into ten country-level monthly series, then each series was converted into a complete monthly time index. Missing values were addressed before model fitting; four forecasting models were trained independently for each country; a grid search was used to select model-specific settings; and final performance was measured on the 2014–2023 holdout period.

This design was appropriate because the study did not only describe historical warming; it also produced a working forecasting artifact that assigns the most suitable model for each ASEAN member state. The artifact therefore connects data preparation, model optimization, and country-level interpretation in one reproducible process.

B. Dataset

The dataset consisted of FAOSTAT Temperature Change on Land monthly country-level records for Brunei Darussalam, Cambodia, Indonesia, Lao PDR, Malaysia, Myanmar, the Philippines, Singapore, Thailand, and Viet Nam from 1961 to 2023 [10], [11]. Each record was organized by country, month, year, and temperature-change value. Table I summarizes the dataset variables and codes used in the analysis.

The ten countries represent different geographic and climatic contexts. Brunei Darussalam, Indonesia, Malaysia, the Philippines, and Singapore are island or coastal equatorial countries, while Cambodia, Lao PDR, Myanmar, Thailand, and Viet Nam are mainland or monsoon-influenced countries. These differences justify country-specific modeling because monthly anomaly behavior can vary according to maritime exposure, monsoon timing, landmass characteristics, and regional climate variability.

Each monthly record was treated as one observation for a specific country and calendar month. The target variable was the temperature anomaly value in degrees Celsius, not absolute air temperature. Positive values indicate warmer-than-baseline conditions, while values near zero indicate conditions close to the reference baseline. This structure makes the data useful for monitoring warming patterns, but also affects the interpretation of percentage-based error metrics.

TABLE I. DATASET VARIABLES AND CODES

Code	Variable	Type	Description
AR	Area	Categorical	ASEAN member state or country identifier
MT	Months	Categorical	Calendar month from January to December
YR	Year	Integer	Observation year from 1961 to 2023
VL	Value	Float	Monthly temperature anomaly in degrees Celsius

C. Preprocessing

Preprocessing removed incomplete country-date identifiers, converted month labels into a monthly datetime index, sorted each country series chronologically, checked for duplicate country-month records, and split the dataset. The final training period covers January 1961 to December 2013, and the final test period covers January 2014 to December 2023. To address missing historical anomaly observations in the Singapore series, a regional and time-aware Ridge Regression imputation procedure was implemented, details of which are described in the following subsection. Missing observations in the final test period were imputed solely to ensure forecasting continuity and were masked from final forecast-accuracy computations.

Missing values were not handled using a single uniform rule because missingness patterns varied by country. Nine country series were complete, whereas Singapore had exactly 360 missing monthly observations out of 756 total months (a 47.6% missing rate). Simple monthly median imputation was avoided because, while it preserves central tendency, it weakens the long-term historical warming trend and inter-annual climate structure that the forecasting models must capture.

Singapore's missing historical values (358 months in the training period and 2 in the test period) were therefore estimated through a regional Ridge Regression model. The model utilized temperature anomalies from neighboring coastal and equatorial countries (Malaysia, Indonesia, and Brunei Darussalam), their regional mean anomaly, a linear year trend, and monthly sine-cosine harmonic indicators to capture seasonality. Crucially, the imputation model was fitted exclusively on observed Singapore data prior to December 2013 to prevent information leakage from the test set. Imputed values in the test set were excluded from final error metric calculations to ensure models were evaluated strictly against actual empirical observations.

D. Forecasting Models

1) Holt-Winters:

To establish a classical statistical baseline, Holt-Winters triple exponential smoothing was configured to capture level, trend, and seasonal components in the monthly anomaly series [12], [13].

Its inclusion was important because it provides a simple and interpretable benchmark for monthly data with recurring annual behavior. The model does not require external predictors, so its performance shows how much accuracy can be achieved using only the level, trend, and seasonality contained in the historical anomaly series.

2) SARIMA:

Seasonal Autoregressive Integrated Moving Average (SARIMA) modeling was implemented to address both non-seasonal dynamics and repeating 12-month seasonal correlations [14].

The seasonal period was fixed at 12 because the records are monthly. SARIMA was retained as a core candidate model because past observations and same-season previous values can contain information about the next monthly anomaly. In the tuned framework, SARIMA orders were selected for each country rather than imposing a single common order across all national series.

3) Prophet:

The study integrated Facebook's Prophet framework, a decomposable additive model tailored for time series with non-linear trends and yearly seasonal cycles [15].

Prophet was relevant because the temperature-anomaly series contains long-term trend movement and recurring yearly structure. Its additive design separates trend and seasonality, which supports interpretation when the objective is not only prediction but also communication of forecast behavior to planning users.

4) XGBoost:

To benchmark non-linear machine learning capabilities against traditional time-series methods, XGBoost was trained using gradient-boosted decision trees [16], [17].

Unlike the statistical models, XGBoost relied on engineered predictors such as lag values, rolling summaries, month indicators, and year trend. This allowed the comparison to test whether a tree-based nonlinear method could improve accuracy over classical time-series approaches. Its country-level performance was interpreted carefully because machine-learning models can underperform when the available features do not capture the climate drivers behind the anomaly changes.

E. Grid Search

Grid search was applied separately for each model and country using rolling validation inside the 1961–2013 training period. The validation folds were trained through 2003 and 2008, then validated on 2004–2008 and 2009–2013. The optimal hyperparameter configuration was determined using the lowest validation MAE. When multiple configurations achieved identical MAE values, RMSE and MASE were considered sequentially to identify the best-performing model. The selected configuration was refitted on the full 1961–2013 training period before final testing on 2014–2023.

The tuning step improved fairness in the comparison because a fixed setting can favor one model while restricting another. Each model was therefore given its own candidate parameter space. Holt-Winters was tuned through trend and seasonal forms, SARIMA through non-seasonal and seasonal orders, Prophet through trend and seasonality controls, and XGBoost through tree and learning parameters.

Rolling validation was used inside the training period so that later observations were predicted from earlier observations. The 2014–2023 test period was not used during tuning. This separation protected the final evaluation from information leakage and ensured that the final comparison reflected performance on unseen monthly observations.

F. Evaluation Metrics

Model accuracy was evaluated using MAE, RMSE, epsilon-adjusted MAPE, sMAPE, and MASE [18]. In (1)–(5), y_i denotes the observed test value, $\hat{y}_i$ denotes the forecast value, $\epsilon = 0.1$, and n denotes the number of evaluated test observations.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1)$$

MAE quantifies the average absolute error in degrees Celsius, providing an intuitive scale where smaller values indicate higher predictive accuracy.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2)$$

RMSE applies quadratic weighting to forecast deviations, penalizing occasional large anomalies more severely than MAE.

$$MAPE_{\epsilon} = \frac{100}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{\max(|y_i|, \epsilon)} \quad (3)$$

MAPE expresses absolute errors relative to observed values; adding an epsilon adjustment ($\epsilon = 0.1$) prevents numeric instability when actual anomalies approach zero.

$$sMAPE = \frac{100}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{(|y_i| + |\hat{y}_i|)/2} \quad (4)$$

The sMAPE metric provides a bounded percentage metric calculated symmetrically over the sum of observed and predicted values, stabilizing evaluation for near-zero anomalies.

$$MASE = \frac{MAE}{\frac{1}{N-m} \sum_{i=m+1}^N |z_i - z_{i-m}|} \quad (5)$$

MASE scales mean absolute error relative to an in-sample seasonal naive benchmark (evaluated at 12-month lag $y_t - y_{t-12}$). Values below 1.0 indicate superior performance relative to a baseline seasonal naive forecast.

Because monthly temperature anomalies can approach zero, MAPE was reported but not used as the primary ranking metric. MAE and RMSE were treated as the primary measures, while sMAPE and MASE supported interpretation.

Regional interpretation was aligned with evidence from Southeast Asian temperature projections [19]. Future diagnostic refinements can use stationarity checks, residual autocorrelation tests, information-criterion selection, and complex-seasonal smoothing extensions [20], [21], [22], [23].

The metrics were interpreted together rather than separately. MAE provides the most direct measure because it reports the average error in degrees Celsius. RMSE highlights models that occasionally produce big mistakes. Symmetric MAPE (sMAPE) and MASE strengthen the evaluation by reducing dependence on the raw magnitude of the anomaly value. This combination gives a more balanced assessment than relying on MAPE alone.

G. Data Analysis

Data analysis followed the study objectives by examining long-term warming patterns, comparing tuned model performance, assigning the best model for each country, and interpreting the planning value of the results. Annual country means described trends, validation errors guided model selection, final test errors supported reporting, and country-level comparisons identified adaptation-relevant implications.

The analysis first summarized the annual mean anomaly pattern for each country to show the long-term temperature-change direction. It then compared the tuned model errors on the holdout period and selected the best model per country, using MAE as the primary criterion. The interpretation connected numerical results with forecast reliability, identifying countries with lower absolute error and countries that require more cautious use of forecasts.

The analytical outputs consisted of aggregate model-performance tables, country-level model assignments, forecast-error visualizations, and adaptation-oriented

interpretation. These outputs were selected because they directly support the study objectives: describing the pattern, evaluating the models, identifying the best model per country, and explaining how the results can guide climate-risk preparedness.

III. RESULTS AND DISCUSSION

A. Long-Term Temperature Pattern

Fig. 1 shows a clear upward pattern in annual mean temperature change across ASEAN from 1961 to 2023. The shared warming signal confirms the regional relevance of the problem, while differences in timing and variability support country-specific modeling. This pattern addressed the need to characterize long-term monthly temperature-change behavior before forecasting.

The upward pattern confirms that the forecasting problem is not limited to short seasonal fluctuation. The models also needed to represent a gradual increase in anomaly values across decades. This finding supports the use of a chronological split because later observations represent a warmer period than most of the training record.

The country-level series also shows that regional warming is shared but not identical across countries. This difference gives practical value to the study because adaptation planning requires national forecasts rather than a pooled regional estimate.

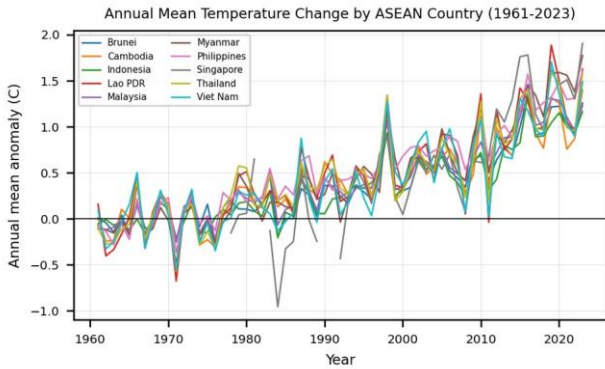


Fig. 1. Annual mean temperature change by ASEAN country from 1961 to 2023.

B. Tuned Model Performance

Table II presents the tuned aggregate test errors averaged across the ten countries. Prophet achieved the lowest MAE, RMSE, sMAPE, and MASE, indicating the strongest overall performance. SARIMA recorded the lowest MAPE (57.20%), but MAPE was not used as the main ranking metric because anomaly values can be close to zero. These results show how the tuned models performed under the same final test period and identify the model with the most accurate regional aggregate forecasts.

Prophet achieved the best aggregate result because it balanced long-term trend estimation with yearly seasonal structure. The result indicates that ASEAN monthly anomaly forecasting benefits from a model that can follow gradual trend movement while still representing annual recurrence.

The small difference between Prophet and SARIMA shows that statistical seasonal structure remains important. SARIMA produced competitive results, but it no longer dominated every country after all models were tuned. This shift strengthens the need for a fair optimization step before drawing conclusions about model superiority.

TABLE II. AGGREGATE FORECAST ERROR ACROSS ASEAN COUNTRIES

Model	MAE	RMSE	MAPE (%)	sMAPE (%)	MASE
Prophet	0.5008	0.6198	60.74	47.54	0.9823
SARIMA	0.5175	0.6346	57.20	51.22	0.9835
XGBoost	0.6293	0.7492	66.56	66.24	1.2454
Holt-Winters	0.6482	0.7648	57.96	76.30	1.1490

C. Country-Specific Model Assignment

Fig. 2 and Table III show the best model for each country using final test MAE as the primary criterion. The result supports country-specific model assignment rather than one model for all ASEAN countries. Prophet was selected for Cambodia, Lao PDR, Myanmar, and Viet Nam; Holt-Winters for Indonesia and the Philippines; SARIMA for Brunei Darussalam and Malaysia; and XGBoost for Singapore and Thailand. Countries with lower MAE values showed more reliable monthly anomaly forecasts, while higher-error mainland countries required more cautious interpretation.

This country-level distribution highlights that distinct climate micro-regions require tailored statistical or machine-learning architectures rather than a single global estimator.

Lower errors were observed in Indonesia, Brunei Darussalam, Malaysia, and the Philippines, where the best-model MAE values were below 0.30 degrees Celsius. Higher errors appeared in Lao PDR, Viet Nam, Thailand, Myanmar, and Singapore, indicating greater forecasting difficulty. These differences show that model choice and forecast reliability must be interpreted at the country level rather than only from aggregate averages.

High MAPE values remained in several mainland countries despite acceptable MAE, RMSE, and MASE behavior. This pattern reflects the weakness of percentage error for anomaly data: when the observed anomaly is near zero, even a moderate absolute error becomes a large percentage error. The result therefore supports the decision to rank models using MAE and to treat MAPE as a secondary diagnostic.

D. Forecast Reliability by Country Group

The country-level error pattern separates the ASEAN series into clearer reliability groups. Indonesia, Brunei Darussalam, Malaysia, and the Philippines formed the strongest group because their selected models produced low MAE values and stable scaled errors. These results indicate that their monthly anomaly patterns were more predictable under the available univariate information.

Cambodia, Thailand, Myanmar, Viet Nam, Lao PDR, and Singapore required more cautious interpretation. Their selected models still provided usable forecasts for several metrics, but the larger MAE or MASE values show that the historical anomaly series alone did not fully explain later test-period movement. For these countries, the forecasts are better treated as early warning inputs than as stand-alone decision rules.

The higher errors in mainland or monsoon-influenced countries show that national climate behavior cannot be reduced to one regional pattern. Temperature anomalies in these countries can be affected by seasonal circulation, rainfall timing, continental heat effects, and year-to-year variability. The model assignments, therefore, give planning agencies a practical signal: the same workflow can be applied regionally, but the chosen model and the confidence placed in its output must remain country-specific.

The results also clarify the role of machine learning in the study. XGBoost was not the best aggregate model, but it was selected for Singapore and Thailand, which means nonlinear lag-based structure can still be useful for selected national series. This prevents the interpretation from favoring one model family in all cases and reinforces the study objective of assigning the best model per country.

The tuned results met the study objectives in a connected way. The upward annual patterns addressed the need to characterize long-term temperature change; the aggregate table showed how the four models performed under the same evaluation setting; the country-level table identified the best model for each country; and the final interpretation translated the error patterns into adaptation-oriented insights. This sequence makes the results useful not only as a model ranking but also as a planning framework.

E. Practical Use of the Forecasts

The forecasts are most appropriate as near-term planning indicators. A monthly anomaly forecast can help agencies anticipate warmer-than-baseline periods and prepare advisories for heat exposure, water demand, energy demand, and agricultural scheduling. The results should be combined with local monitoring because the models estimate expected anomaly behavior rather than exact future weather conditions.

For countries with lower errors, the forecasts can support routine monitoring and early planning discussions. For countries with higher errors, the forecast output should be accompanied by uncertainty communication and additional evidence from climate agencies. This distinction is important because the value of the framework is not only in producing forecasts but also in showing where those forecasts are more reliable and where further model development is needed.

TABLE III. BEST MODEL PER COUNTRY ON THE 2014–2023 TEST PERIOD

Country	Model	MAE	RMSE	MAPE (%)	sMAPE (%)	MASE
Brunei Darussalam	SARIMA	0.2582	0.3418	37.82	26.79	0.6394
Cambodia	Prophet	0.4985	0.6324	87.77	54.62	0.8367
Indonesia	Holt-Winters	0.2259	0.2985	23.59	23.54	0.7751
Lao PDR	Prophet	0.7143	0.9056	101.64	64.80	0.8733
Malaysia	SARIMA	0.2614	0.3350	26.79	24.35	0.7641
Myanmar	Prophet	0.5520	0.6584	49.69	45.76	1.0329
Philippines	Holt-Winters	0.2844	0.3537	31.93	23.50	0.7576
Singapore	XGBoost	0.5013	0.6414	42.87	37.51	1.0941
Thailand	XGBoost	0.5782	0.7288	86.08	60.08	0.8780
Viet Nam	Prophet	0.6402	0.7884	94.76	67.01	0.8346

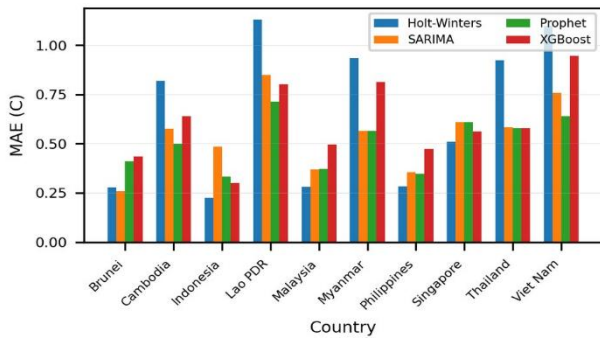


Fig. 2. Per-country MAE by tuned forecasting model for the 2014–2023 test period.

F. Adaptation-Oriented Insights

The empirical model assignments convert historical anomaly records into actionable decision support for regional climate adaptation. For maritime states with lower absolute errors (Indonesia, Brunei Darussalam, Malaysia, and the Philippines), monthly forecast signals provide reliable inputs for agricultural scheduling, extreme heat exposure alerts, and energy grid load monitoring.

Conversely, mainland countries facing higher forecast errors (Lao PDR, Viet Nam, Thailand, and Myanmar) require a more conservative operational posture, where univariate projections are supplemented by exogenous climate drivers prior to policy integration.

Ultimately, these findings demonstrate that ASEAN climate services can establish a unified regional evaluation pipeline while maintaining country-specific model assignments tailored to national anomaly dynamics.

IV. CONCLUSION

This study established a comparative, data-driven forecasting framework tailored to monthly temperature anomalies across the ten ASEAN member states. While Prophet demonstrated the top aggregate accuracy across the region after grid-search tuning, individual country evaluations confirmed that optimal model selection varies significantly by geographical and climatic context.

The study was limited by its reliance on univariate monthly anomaly records and the absence of external climate drivers, including ENSO indicators, rainfall, humidity, sea-surface temperature, land-use change, and monsoon-cycle information. Singapore also required special treatment because the original source contained substantial missing observations. These limitations mean that the results serve as a forecasting baseline for planning support rather than a complete operational climate-risk system.

Future work should integrate exogenous climate predictors, test hybrid statistical and machine-learning models, compare alternative imputation strategies, and evaluate longer rolling forecast horizons. The framework should also be tested on higher-resolution gridded climate products and expanded with uncertainty intervals. These directions would strengthen country-level forecast reliability and improve the value of the framework for ASEAN climate-risk preparedness.

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